Source of multiplexed heralded down-converted entangled photon pairs — In search for pairs on demand

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Can we go around Kok-Braunstein's interdiction: "A maximally entangled state cannot be generated conditioned on detection of two auxiliary photons." Śliwa and Banaszek conclude that "this would rule out using four photons, to produce maximally entangled pairs by means of conditional detection."

Apparently, yes! Kok and Braunstein obtained their result under the following assumption: "every detector needs to detect at most one photon" and if we found a way to preselect photon pairs based on detecting of more than one photon by some detectors we would be able to go around Śliwa and Banaszek's conclusion. This is exactly what we did in our design below by making use of photon number discriminating (TES) detectors.

Signal (H) and idler (V) photons can take four different paths after being split at the first polarizing beam splitter (PBS). Possible combinations for them to end up together or separated in one of the four paths (*i*-*iv*) are:

- Top: (a) H-i, V-i; (b) H-i, V-iii; (c) H-iii, V-i; (d) H-iii, V-iii;
- Bottom: (e) H-ii, V-ii; (f) H-ii, V-iv; (g) H-iv, V-ii; (h) H-iv, V-iv.

	(a)	(b)	(c)	(d)
(e)	4	3	3	⊗
(f)	3	$2\times H$	✓	1
(g)	3	✓	$2\times V$	1
(h)	⊗	1	1	0

Acceptable (✓) & excluded combinations of photons as measured by detectors d$_i$; ⊗ = both pairs coming from a single cavity.

✓: $|\Psi^+\rangle=\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_2+|V\rangle_1|H\rangle_2)\xrightarrow{BS_1}\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_1-|H\rangle_2|V\rangle_2)\xrightarrow{BS_2}\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_2-|V\rangle_1|H\rangle_2)=|\Psi^-\rangle\to$ d$_i$ out →

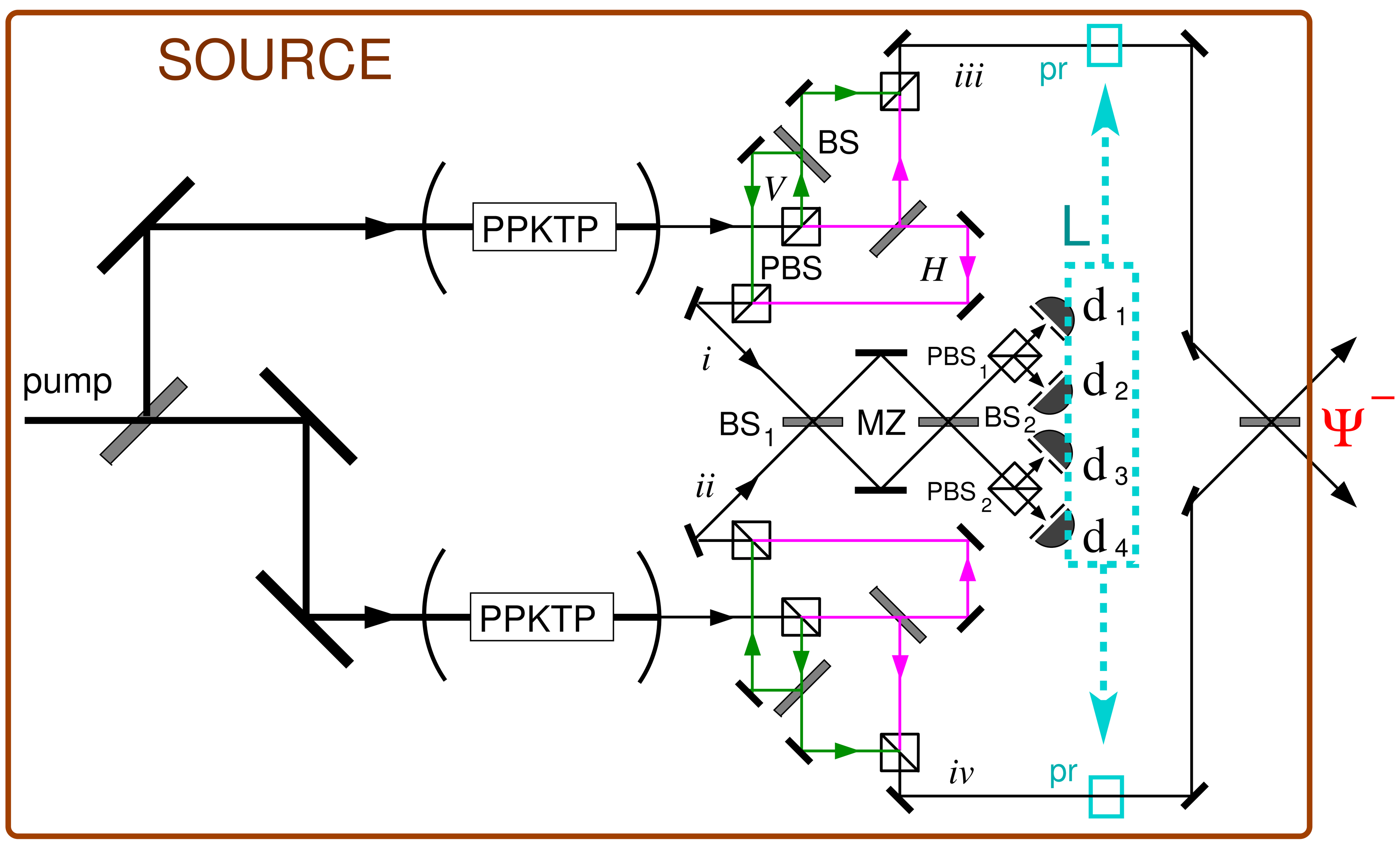
⊗: $\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_1+|H\rangle_2|V\rangle_2)\xrightarrow{BS_1}\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_1+|H\rangle_2|V\rangle_2)\xrightarrow{BS_2}\frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_1+|H\rangle_2|V\rangle_2)\to$ d$_i$s block outputs

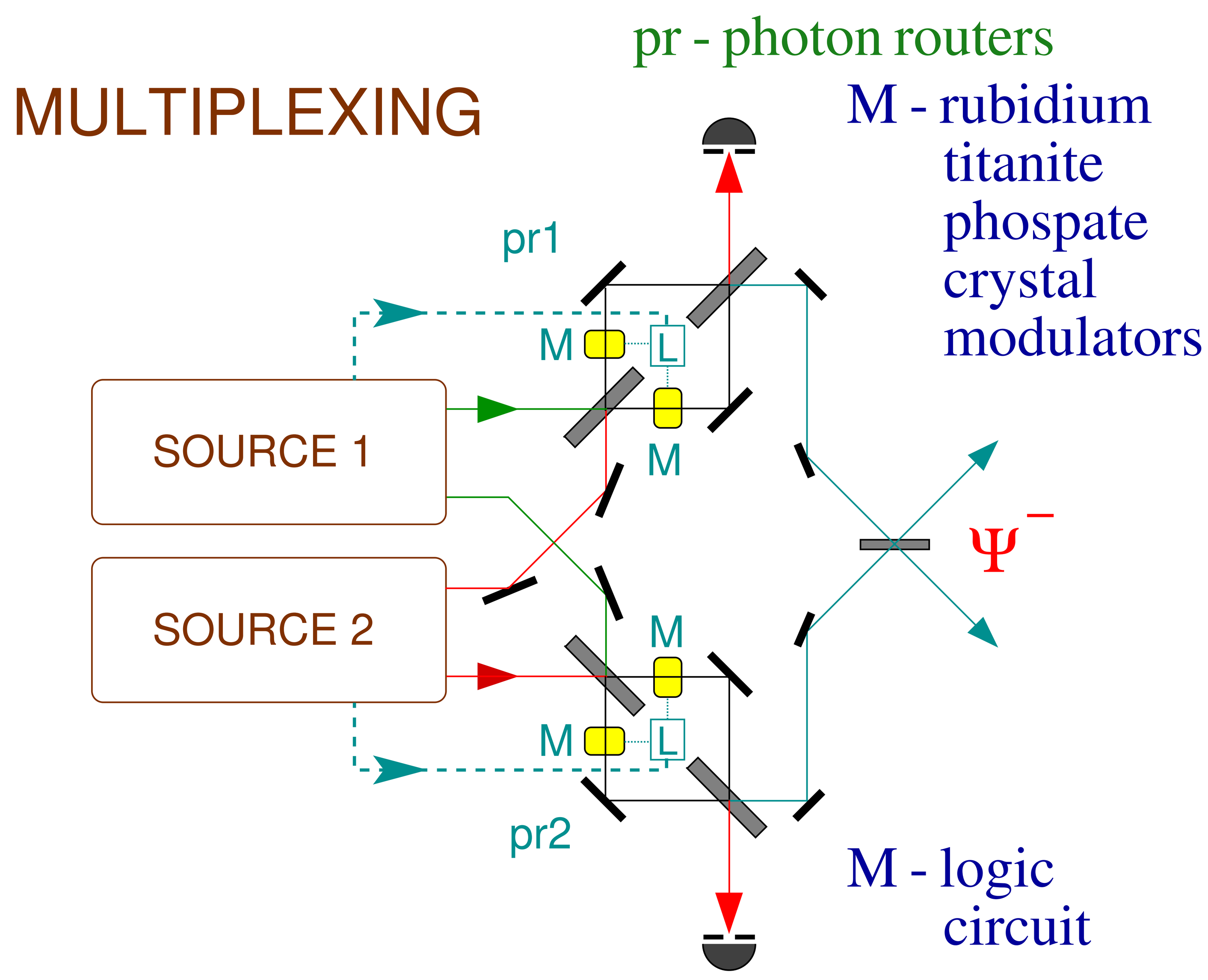
etc. for other combinations of photons from two pairs .... and except for 4 photons from 3 pairs below ....

	2/2	3/2	2/3	3/3	4/3	5/3	6/3
⊗	*vac* 12.5%	1 p 6.2%	*vac* 12.5%	1 p 6.2%	2 p 18.7%	3 p 25%	4 p 3.1%
$\lvert\Psi^-\rangle$	-	-	-	-	✓ 25%	-	-

vac stands for *vacuum* and p for photon(s); e.g., 4/3 means for 4 photons from 3 pairs 25% of 18.7% of them will be falsely sent out; all other unwanted combinations will be blocked.

Efficiency of our source. The efficiency of obtaining a pair in a PPKTP crystal in a cavity is up to 80%. For two such sources $0.8^2=0.64$. So, $1-0.64=0.36$ is the probability of obtaining unwanted photons. Taking into account that the probabilities with which 1 or 3 pairs are down-converted in a Poissonian distribution is ca. 5%, we get, after taking out all the cases eliminated by detectors d$_1$-d$_4$, that the ouput state $|\Psi^-\rangle$ will be generated with the probability of 98%. When we include the efficiency of the TES detectors of 98%, we obtain that the overall efficiency of our source is $0.98\times0.98^3=0.922$, i.e., ca. 92%, since on everage (less than) ca. 3 detectors are required to fire simultaneously.





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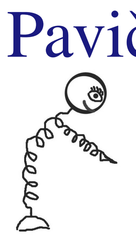

Resarch from a parallel universe: Pavičić, M., Quantum Contextuality, *Quantum*, 7, 953-1-68 (2023)